Topic: Laplace transforms for initial value problems

Question: Find the solution to the second order equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' + 3y' + 2y = t$$

Answer choices:

A $y(t) = \frac{1}{2}t - \frac{3}{4} + e^{t} - \frac{1}{4}e^{2t}$

B $y(t) = \frac{1}{2}t - \frac{3}{4} + e^{2t} - \frac{1}{4}e^{-t}$

C $y(t) = \frac{1}{2}t - \frac{3}{4} + e^{-t} - \frac{1}{4}e^{-2t}$

D $y(t) = \frac{1}{2}t - \frac{3}{4} + e^{2t} - \frac{1}{4}e^{2t}$

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Solution: C

From our table of Laplace transforms, we know

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

Plugging these transforms into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 3(sY(s) - y(0)) + 2Y(s) = \frac{1}{s^2}$$

Now we'll plug in the initial conditions $y(0) = 0$ and $y'(0) = 0$ in order to simplify the transform.

$$s^2Y(s) - s(0) - 0 + 3(sY(s) - 0) + 2Y(s) = \frac{1}{s^2}$$

$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{1}{s^2}$$

Factor out $Y(s)$, then isolate it on the left side of the equation.

$$Y(s)(s^2 + 3s + 2) = \frac{1}{s^2}$$

$$Y(s) = \frac{1}{s^2(s^2 + 3s + 2)}$$

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$$Y(s) = \frac{1}{s^2(s+2)(s+1)}$$

We'll need to use a partial fractions decomposition.

$$\frac{1}{s^2(s+2)(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{s+1}$$

To solve for B, we'll remove the s^2 factor, set $s = 0$, and find the value of the left side.

$$\frac{1}{(s+2)(s+1)}$$

$$\frac{1}{(0+2)(0+1)} = \frac{1}{2(1)} = \frac{1}{2}$$

To solve for C, we'll remove the $(s+2)$ factor, set $s = -2$, and find the value of the left side.

$$\frac{1}{s^2(s+1)}$$

$$\frac{1}{(-2)^2(-2+1)} = \frac{1}{4(-1)} = -\frac{1}{4}$$

To solve for D, we'll remove the $(s+1)$ factor, set $s = -1$, and find the value of the left side.

$$\frac{1}{s^2(s+2)}$$

$$\frac{1}{(-1)^2(-1+2)} = \frac{1}{1(1)} = 1$$

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Now that we have the values of B, C, and D, we can plug those into the decomposition,

$$\frac{1}{s^2(s+2)(s+1)} = \frac{A}{s} + \frac{\frac{1}{2}}{s^2} + \frac{-\frac{1}{4}}{s+2} + \frac{1}{s+1}$$

and then pick a value we haven't used for s yet ($s \neq -2,\ -1,\ 0$), like $s = 1$, plug it in, and solve for A.

$$\frac{1}{1^2(1+2)(1+1)} = \frac{A}{1} + \frac{\frac{1}{2}}{1^2} + \frac{-\frac{1}{4}}{1+2} + \frac{1}{1+1}$$

$$\frac{1}{3(2)} = A + \frac{1}{2} - \frac{1}{12} + \frac{1}{2}$$

$$A = \frac{1}{6} + \frac{1}{12} - \frac{1}{2} - \frac{1}{2}$$

$$A = \frac{2}{12} + \frac{1}{12} - \frac{6}{12} - \frac{6}{12}$$

$$A = -\frac{9}{12}$$

$$A = -\frac{3}{4}$$

Plugging the values we found for A, B, C, and D back into the partial fractions decomposition gives

$$Y(s) = \frac{-\frac{3}{4}}{s} + \frac{\frac{1}{2}}{s^2} + \frac{-\frac{1}{4}}{s+2} + \frac{1}{s+1}$$

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and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$Y(s) = -\frac{3}{4}\left(\frac{1}{s}\right) + \frac{1}{2}\left(\frac{1}{s^2}\right) - \frac{1}{4}\left(\frac{1}{s-(-2)}\right) + \frac{1}{s-(-1)}$$

The terms remaining inside the parentheses should remind us of the transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(e^{-2t}) = \frac{1}{s-(-2)} \qquad \mathscr{L}(e^{-t}) = \frac{1}{s-(-1)}$$

So we'll make these substitutions to put the equation back in terms of the original variable t, instead of the transform variable s, which will give us the solution to the second order nonhomogeneous differential equation.

$$y(t) = -\frac{3}{4}(1) + \frac{1}{2}(t) - \frac{1}{4}e^{-2t} + e^{-t}$$

$$y(t) = -\frac{3}{4} + \frac{1}{2}t - \frac{1}{4}e^{-2t} + e^{-t}$$

$$y(t) = \frac{1}{2}t - \frac{3}{4} + e^{-t} - \frac{1}{4}e^{-2t}$$

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Topic: Laplace transforms for initial value problems

Question: Find the solution to the second order equation, given $y(0) = 4$ and $y'(0) = 9$.

$$y''(t) - 3y'(t) + 2y(t) = 4e^{3t}$$

Answer choices:

A $y(t) = 2e^{3t} + e^{2t} + e^{t}$

B $y(t) = e^{3t} + 2e^{2t} + e^{t}$

C $y(t) = e^{4t} + 2e^{3t} + e^{2t}$

D $y(t) = 2e^{3t} + 2e^{2t} + e^{t}$

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Solution: A

From our table of Laplace transforms, we know

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(e^{at}) = \frac{1}{s-a}$$

Plugging these transforms into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - 3(sY(s) - y(0)) + 2Y(s) = 4\left(\frac{1}{s-3}\right)$$

Now we'll plug in the initial conditions $y(0) = 4$ and $y'(0) = 9$ in order to simplify the transform.

$$s^2Y(s) - s(4) - 9 - 3(sY(s) - 4) + 2Y(s) = 4\left(\frac{1}{s-3}\right)$$

$$s^2Y(s) - 4s - 9 - 3sY(s) + 12 + 2Y(s) = \frac{4}{s-3}$$

Solve for $Y(s)$ by collecting all the $Y(s)$ terms on one side, and moving all other terms to the other side.

$$s^2Y(s) - 3sY(s) + 2Y(s) = \frac{4}{s-3} + 4s - 3$$

Factor out $Y(s)$, then isolate it on the left side of the equation.

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$$Y(s)(s^2-3s+2)=\frac{4}{s-3}+4s-3$$

$$Y(s)(s^2-3s+2)=\frac{4}{s-3}+\frac{(4s-3)(s-3)}{s-3}$$

$$Y(s)(s^2-3s+2)=\frac{4s^2-15s+13}{s-3}$$

$$Y(s)=\frac{4s^2-15s+13}{(s-3)(s-2)(s-1)}$$

We'll need to use a partial fractions decomposition.

$$\frac{4s^2-15s+13}{(s-3)(s-2)(s-1)}=\frac{A}{s-3}+\frac{B}{s-2}+\frac{C}{s-1}$$

To solve for A, we'll remove the $(s-3)$ factor, set $s=3$, and find the value of the left side.

$$\frac{4s^2-15s+13}{(s-2)(s-1)}$$

$$\frac{4(3)^2-15(3)+13}{(3-2)(3-1)}=\frac{4(9)-45+13}{1(2)}=\frac{4}{2}=2$$

To solve for B, we'll remove the $(s-2)$ factor, set $s=2$, and find the value of the left side.

$$\frac{4s^2-15s+13}{(s-3)(s-1)}$$

$$\frac{4(2)^2-15(2)+13}{(2-3)(2-1)}=\frac{4(4)-30+13}{-1(1)}=\frac{-1}{-1}=1$$

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To solve for C, we'll remove the $(s-1)$ factor, set $s=1$, and find the value of the left side.

$$\frac{4s^2-15s+13}{(s-3)(s-2)}$$

$$\frac{4(1)^2-15(1)+13}{(1-3)(1-2)}=\frac{4(1)-15+13}{-2(-1)}=\frac{2}{2}=1$$

Plugging the values we found for A, B, and C back into the partial fractions decomposition gives

$$Y(s)=\frac{2}{s-3}+\frac{1}{s-2}+\frac{1}{s-1}$$

and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$Y(s)=2\left(\frac{1}{s-3}\right)+\frac{1}{s-2}+\frac{1}{s-1}$$

The terms remaining should remind us of the transforms

$$\mathscr{L}(e^{3t})=\frac{1}{s-3} \qquad \mathscr{L}(e^{2t})=\frac{1}{s-2} \qquad \mathscr{L}(e^{t})=\frac{1}{s-1}$$

So we'll make these substitutions to put the equation back in terms of the original variable t, instead of the transform variable s, which will give us the solution to the second order nonhomogeneous differential equation.

$$y(t)=2e^{3t}+e^{2t}+e^{t}$$

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Topic: Laplace transforms for initial value problems

Question: Find the solution to the second order equation, given $y(0) = 1$ and $y'(0) = 2$.

$$y'' - 5y' + 6y = 3t$$

Answer choices:

A $y(t) = \frac{5}{12} + \frac{1}{2}t + \frac{1}{4}e^{2t} - \frac{1}{3}e^{3t}$

B $y(t) = \frac{5}{12} + \frac{1}{2}t + \frac{1}{4}e^{2t} + \frac{1}{3}e^{3t}$

C $y(t) = \frac{5}{12} + \frac{1}{2}t - \frac{1}{12}e^{t}$

D $y(t) = \frac{5}{12} + \frac{1}{2}t + \frac{7}{12}e^{5t}$

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Solution: B

From our table of Laplace transforms, we know

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y') = sY(s) - y(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(t) = \frac{1}{s^2}$$

Plugging these transforms into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) - 5(sY(s) - y(0)) + 6Y(s) = 3\left(\frac{1}{s^2}\right)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = 2$ in order to simplify the transform.

$$s^2Y(s) - s(1) - 2 - 5(sY(s) - 1) + 6Y(s) = 3\left(\frac{1}{s^2}\right)$$

$$s^2Y(s) - s - 2 - 5sY(s) + 5 + 6Y(s) = \frac{3}{s^2}$$

Solve for $Y(s)$ by collecting all the $Y(s)$ terms on one side, and moving all other terms to the other side.

$$s^2Y(s) - 5sY(s) + 6Y(s) = \frac{3}{s^2} + s - 3$$

Factor out $Y(s)$, then isolate it on the left side of the equation.

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$$Y(s)(s^2-5s+6)=\frac{3}{s^2}+s-3$$

$$Y(s)(s^2-5s+6)=\frac{3}{s^2}+\frac{s^2(s-3)}{s^2}$$

$$Y(s)(s^2-5s+6)=\frac{s^3-3s^2+3}{s^2}$$

$$Y(s)=\frac{s^3-3s^2+3}{s^2(s-3)(s-2)}$$

We'll need to use a partial fractions decomposition.

$$\frac{s^3-3s^2+3}{s^2(s-3)(s-2)}=\frac{A}{s}+\frac{B}{s^2}+\frac{C}{s-3}+\frac{D}{s-2}$$

To solve for B, we'll remove the s^2 factor, set $s=0$, and find the value of the left side.

$$\frac{s^3-3s^2+3}{(s-3)(s-2)}$$

$$\frac{0^3-3(0)^2+3}{(0-3)(0-2)}=\frac{3}{-3(-2)}=\frac{3}{6}=\frac{1}{2}$$

To solve for C, we'll remove the $(s-3)$ factor, set $s=3$, and find the value of the left side.

$$\frac{s^3-3s^2+3}{s^2(s-2)}$$

$$\frac{3^3-3(3)^2+3}{3^2(3-2)}=\frac{27-3(9)+3}{9(1)}=\frac{3}{9}=\frac{1}{3}$$

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To solve for D, we'll remove the $(s-2)$ factor, set $s=2$, and find the value of the left side.

$$\frac{s^3-3s^2+3}{s^2(s-3)}$$

$$\frac{2^3-3(2)^2+3}{2^2(2-3)}=\frac{8-3(4)+3}{4(-1)}=\frac{-1}{-4}=\frac{1}{4}$$

Now that we have the values of B, C, and D, we can plug those into the decomposition,

$$\frac{s^3-3s^2+3}{s^2(s-3)(s-2)}=\frac{A}{s}+\frac{\frac{1}{2}}{s^2}+\frac{\frac{1}{3}}{s-3}+\frac{\frac{1}{4}}{s-2}$$

and then pick a value we haven't used for s yet ($s\neq 0,\ 2,\ 3$), like $s=1$, plug it in, and solve for A.

$$\frac{1^3-3(1)^2+3}{1^2(1-3)(1-2)}=\frac{A}{1}+\frac{\frac{1}{2}}{1^2}+\frac{\frac{1}{3}}{1-3}+\frac{\frac{1}{4}}{1-2}$$

$$\frac{1-3+3}{-2(-1)}=A+\frac{1}{2}+\frac{\frac{1}{3}}{-2}+\frac{\frac{1}{4}}{-1}$$

$$\frac{1}{2}=A+\frac{1}{2}-\frac{1}{6}-\frac{1}{4}$$

$$A=\frac{1}{2}-\frac{1}{2}+\frac{1}{6}+\frac{1}{4}$$

$$A=\frac{4}{24}+\frac{6}{24}$$

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$$A = \frac{5}{12}$$

Plugging the values we found for A, B, C, and D back into the partial fractions decomposition gives

$$Y(s) = \frac{\frac{5}{12}}{s} + \frac{\frac{1}{2}}{s^2} + \frac{\frac{1}{3}}{s-3} + \frac{\frac{1}{4}}{s-2}$$

and then we can rearrange each term in the decomposition to make it easier to find a matching formula in the Laplace transform table.

$$Y(s) = \frac{5}{12}\left(\frac{1}{s}\right) + \frac{1}{2}\left(\frac{1}{s^2}\right) + \frac{1}{3}\left(\frac{1}{s-3}\right) + \frac{1}{4}\left(\frac{1}{s-2}\right)$$

The terms remaining inside the parentheses should remind us of the transforms

$$\mathscr{L}(1) = \frac{1}{s} \qquad \mathscr{L}(t) = \frac{1}{s^2}$$

$$\mathscr{L}(e^{3t}) = \frac{1}{s-3} \qquad \mathscr{L}(e^{2t}) = \frac{1}{s-2}$$

So we'll make these substitutions to put the equation back in terms of the original variable t, instead of the transform variable s, which will give us the solution to the second order nonhomogeneous differential equation.

$$y(t) = \frac{5}{12}(1) + \frac{1}{2}(t) + \frac{1}{3}e^{3t} + \frac{1}{4}e^{2t}$$

$$y(t) = \frac{5}{12} + \frac{1}{2}t + \frac{1}{3}e^{3t} + \frac{1}{4}e^{2t}$$

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$$y(t) = \frac{5}{12} + \frac{1}{2}t + \frac{1}{4}e^{2t} + \frac{1}{3}e^{3t}$$

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